

Multiple Choice

1. **Answer:** B

Options: A) $O(1)$; B) $O(N)$; C) $O(\log N)$; D) $O(N^2)$

Note: The correct answer is $O(N)$ because the nearest neighbors algorithm requires a linear search through all training instances to find the closest neighbors for classification, resulting in a runtime that scales directly with the number of instances N in the dataset.

2. **Answer:** C

Options: A) Regularization is too low and model is overfitting; B) Regularization is too high and model is underfitting; C) Step size is too large; D) Step size is too small

Note: A step size that is too large can cause the model to overshoot the optimal solution during training, leading to increased training loss over epochs as it fails to converge.

3. **Answer:** B

Options: A) Increase the amount of training data.; B) Improve the optimisation algorithm being used for error minimisation.; C) Decrease the model complexity.; D) Reduce the noise in the training data.

Note: Improving the optimization algorithm primarily affects the convergence speed and efficiency of training rather than directly addressing the model's complexity and its tendency to overfit the training data.

4. **Answer:** D

Options: A) It relates inputs to outputs.; B) It is used for prediction.; C) It may be used for interpretation.; D) It discovers causal relationships

Note: The statement is false because regression analysis identifies statistical relationships between variables but does not establish causation, as correlation does not imply causation.

5. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Industrial-scale neural networks are typically trained on GPUs due to their superior parallel processing capabilities, and the ResNet-50 model actually has around 25 million parameters, not over 1 billion.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Clustering is effective for detecting fraudulent credit card transactions because

it identifies and groups similar spending behaviors, allowing for the identification of anomalies that may indicate fraudulent activity.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the joint probability distribution $P(X, Y, Z)$ can only be expressed as the product of the marginal distributions $P(X)$, $P(Y)$, and $P(Z)$ if X , Y , and Z are independent random variables, which is not generally the case.

8. **Answer:** True

Options: A) True; B) False

Note: Out-of-distribution detection is considered a form of anomaly detection because both involve identifying data points that significantly deviate from the expected distribution, indicating they are unusual or not representative of the normal dataset.

9. **Answer:** True

Options: A) True; B) False

Note: A smaller error rate implies that the model is performing better, requiring a larger sample size to reliably distinguish between the subtle differences in performance and to ensure that the observed results are not due to random chance.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because a null space dimensionality of 2 indicates that there are two linearly independent vectors that span the null space of the matrix A , implying that the rank of A is reduced by two dimensions in relation to its total number of columns.

Long Form Response

11. **Answer:** `def average_tuple(nums): | return result`

Note: correct because it provides a function definition that indicates the intention to compute the average of numbers contained in a tuple of tuples, which is essential for solving the posed problem in the context of data processing in machine learning.

12. **Answer:** `def is_key_present(d,x): | return True | return False`

Note: correct because it defines a function that checks for the presence of a specified key in a dictionary and appropriately returns a Boolean value indicating its existence.

End of Answer Key